

Run UVM on Verilator

Modules twenty-two and twenty-three moved you from host literacy to Makefile knobs

Verilator-first offline run

- Browser sketches taught factory, agents, and sequences, they do not compile uvm_pkg
- Offline you use the in-course Makefile: export UVM home, make run with Verilator
- The flow compiles uvm_pkg plus DUT plus test into a C++ executable
- First cold compile can take many minutes; later rebuilds reuse objects

Offline workflow

- Open examples verilator-uvm-hello beside this curriculum
- Export UVM home to Accellera src
- Run make dry-run if you skipped module twenty-three
- Then make run
- Watch Verilator compile stages finish, then the binary print UVM info and a report summary
- Capture the exact command and the pass or fail line

Real Verilator UVM run

```
wsl - module24-offline-uvm-verilator-run/examples/navigation

# Track A - real Unix

== pwd ==
/mnt/d/proj/designs/learning/courses/learn_uvm2017/examples/verilator-uvm-hello
total 8
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 ..
-rwxrwxrwx 1 yongfu yongfu 2289 Jul 21 10:41 Makefile
-rwxrwxrwx 1 yongfu yongfu 1708 Jul 21 10:46 README.md
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 dut
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 scripts
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 tb
dut:
and_gate.v

tb:
test_and_gate_uvm.sv
== Makefile head ==
# Minimal Accellera UVM 2017 + Verilator hello (and-gate)
#
# Prerequisites:
# - Verilator 5.036+ (5.042+ recommended) on PATH
# - make + C++ toolchain
# - Accellera UVM 2017-1.0 source tree + set UVM_HOME to its src/ dir
#
# Example (after extracting the Accellera tarball):
# export UVM_HOME=/path/to/1800.2-2017-1.0/src
# make run SIM=verilator
#
# Optional monorepo convenience (archive fidelity only - not required):
# export UVM_HOME=$PWD/../../learn_uvm2017_sv_verilator/tools/uvm-2017/1800.2-2017-1.0/src

SIM ?= verilator
TEST ?= test_and_gate_uvm
TOplevel ?= $(TEST)
# UVM test class registered inside tb/test_and_gate_uvm.sv
UVM_TESTNAME ?= AndGateTest

ifeq ($(UVM_HOME),)
$(error UVM_HOME is not set. Point it at Accellera UVM 2017 ./src (see README.md))
endif

UVM_SRC_DIR = $(if $(wildcard $(UVM_HOME)/src/uvm_pkg.sv),$(UVM_HOME)/src,$(UVM_HOME))
INCLUDES = -I$(UVM_SRC_DIR) +incdir+$(UVM_SRC_DIR) +define+UVM_NO_DPI
VERILATOR_FLAGS = -sv --timing --trace -Wno-fatal -Wno-TIMESCALEMOD -Wno-WIDTHTRUNC

DUT_SRC = dut/and_gate.v
TEST_SRC = tb/$(TEST).sv

.PHONY: compile run clean dry-run

ifeq ($(SIM),verilator)
compile:
@echo "Compiling $(TEST) with Verilator + UVM from $(UVM_SRC_DIR)"
verilator $(VERILATOR_FLAGS) $(INCLUDES) \
--binary \
$(UVM_SRC_DIR)/uvm_pkg.sv \
UVM_HOME=unset
(Full make run omitted in demo capture - learners run: make run SIM=verilator)
```

Real shell — Verilator UVM make run

Real Verilator UVM run — try these

```
# cd in-course Verilator + UVM hello
cd courses/learn_uvm2017/examples/verilator-uvm-hello

# export UVM_HOME — Accellera UVM 2017 src/
export UVM_HOME=/path/to/1800.2-2017-1.0/src

# make run — Verilator compile + execute +UVM_TESTNAME=AndGateTest
make run SIM=verilator
```

Pitfalls to watch

- Unset UVM home fails before useful compile output
- A browser UVM sketch pass is not a Verilator UVM pass
- Patience on first compile, UVM templates are heavy
- If Verilator is too old for UVM twenty seventeen, upgrade or note the version blocker
- Keep the ignored legacy archive out of your default path

Your turn

- Complete the checklist: dry-run green, make run finished, pass or fail recorded
- Prefer the in-course and-gate hello unless your site policy points elsewhere
- When you are ready, take the short quiz, then continue to the wrap module